

ECON 6343: Econometrics III

Problem Set 3 - GEV Models

Elizabeth Adusei

Question 1: Multinomial Logit Model

I estimated a multinomial logit model of occupational choice using the individual-specific covariates age, white, and collgrad, together with occupation-specific expected log wages. Occupation 8 (Other) is the normalized alternative.

$$X_i = (\text{age}_i, \text{white}_i, \text{collgrad}_i)$$

$$V_{ij} = X_i \beta_j + \gamma (Z_{ij} - Z_{i8}), j = 1, \dots, 7$$

The estimated coefficient vector is organized below. The final coefficient is the common relative expected log-wage coefficient, gamma.

Occupation	age	white	collgrad
1	0.0557	0.0834	-2.3449
2	0.0450	0.7366	-3.1532
3	0.0926	-0.0842	-4.2733
4	0.0239	0.7231	-3.7494
5	0.0361	-0.6438	-4.2797
6	0.0853	-1.1714	-6.6787
7	0.0866	-0.7979	-4.9691
8 (normalized)	0	0	0

Estimated relative expected log-wage coefficient: $\gamma\text{-hat} = -0.0942$

Question 2: Interpretation of gamma-hat

The parameter gamma measures how the relative expected log wage, $(Z_{ij} - Z_{i8})$, affects the utility of choosing occupation j relative to the normalized occupation 8.

$$\gamma\text{-hat} = -0.0942$$

Holding the individual characteristics and other variables fixed, an increase in occupation j's expected log wage relative to occupation 8 decreases the utility index for occupation j and therefore lowers its choice probability relative to occupation 8. The negative sign is somewhat counterintuitive because a higher expected wage would ordinarily be expected to make an occupation more attractive. The value -0.0942 is a coefficient, not a probability; the multinomial-logit probabilities remain between zero and one.

Question 3: Nested Logit Model

The nested logit model uses the assignment's nesting structure:

White Collar (WC) = {1, 2, 3} | Blue Collar (BC) = {4, 5, 6, 7} | Other = {8}

The estimated parameters are organized below.

Parameter group	age	white	collgrad
beta_WC	0.0983	0.1059	-3.0829
beta_BC	0.1117	-0.7883	-4.3520

λ_{WC} = -0.2841

λ_{BC} = -0.2472

γ = 0.6723

The estimated nesting parameters are negative. The optimization routine used for the exercise is unconstrained LBFGS, so the estimation does not restrict the lambda parameters to a particular interval. I therefore report the estimates produced by the specified estimation procedure.

Question 4: Main Wrapper Function

The estimation procedure is placed inside the `allwrap()` function. It loads the data, estimates the multinomial logit model, estimates the nested logit model, and prints the resulting parameter estimates. The script file reads the source file and calls `allwrap()` to execute the complete procedure.

Question 5: Unit Tests

A separate test script reads the source file and tests the main components of the program. All eight tests passed.

Test	Passed	Total
Data loading	4	4
Multinomial logit likelihood	2	2
Nested logit likelihood	2	2
Total	8	8

Submitted Julia files

PS3_Adusei_source.jl

PS3_Adusei_script.jl

PS3_Adusei_tests.jl